

Recursion calls for positive power fn:-

$$2^{10}, n > 0$$

$$\text{half} = 10/2 = 5$$

$$\text{halfPow} = \text{pow}(2, 5)$$

$$\text{return } 32 \times 32 = 1024$$

Returned as the final fn output.

$$2^5, n > 0$$

$$\text{half} = 5/2 = 2$$

$$\text{halfPow} = \text{pow}(2, 2)$$

$$\text{return } 2 \times 4 \times 4 = 32$$

$$2^2, n > 0$$

$$\text{half} = 2/2 = 1$$

$$\text{halfPow} = \text{pow}(2, 1)$$

$$\text{return } 2 \times 2 = 4$$

$$2^1, n > 0$$

$$\text{half} = 1/2 = 0$$

$$\text{halfPow} = \text{pow}(2, 0)$$

$$\text{return } 2 \times 1 \times 1 = 2$$

$$2^0, n = 0$$

$$\text{return } 1$$

My Pow :-

$$x = 2, n = 10$$

We need to find 2^{10}

If we carefully observe,

$$2^{10} = 2^5 \times 2^5$$

$$2^5 = 2^2 \times 2^2 \times 2$$

$$2^2 = 2 \times 2$$

$$2 = 2$$

So, we can find the power more efficiently by using **divide and conquer**.

→ Just find $x^{n/2}$ and multiply it by itself, if n is even, otherwise also multiply x with it.

Basically,

$$n \text{ is even} \rightarrow x^{n/2} \times x^{n/2}$$

$$n \text{ is odd} \rightarrow x^{n/2} \times x^{n/2} \times x$$

→ Now repeat the above process for smaller values of n

If ' n ' is negative, just do $1/(-n)$, as $x^{-n} = 1/x^n$

Now, there is an **edge case**,

If $n = \text{Integer MIN-VALUE}$ (-2147483648), it can't be converted to $-ve$.

∴ Take a helper method with **long n** as parameter and everything will work.